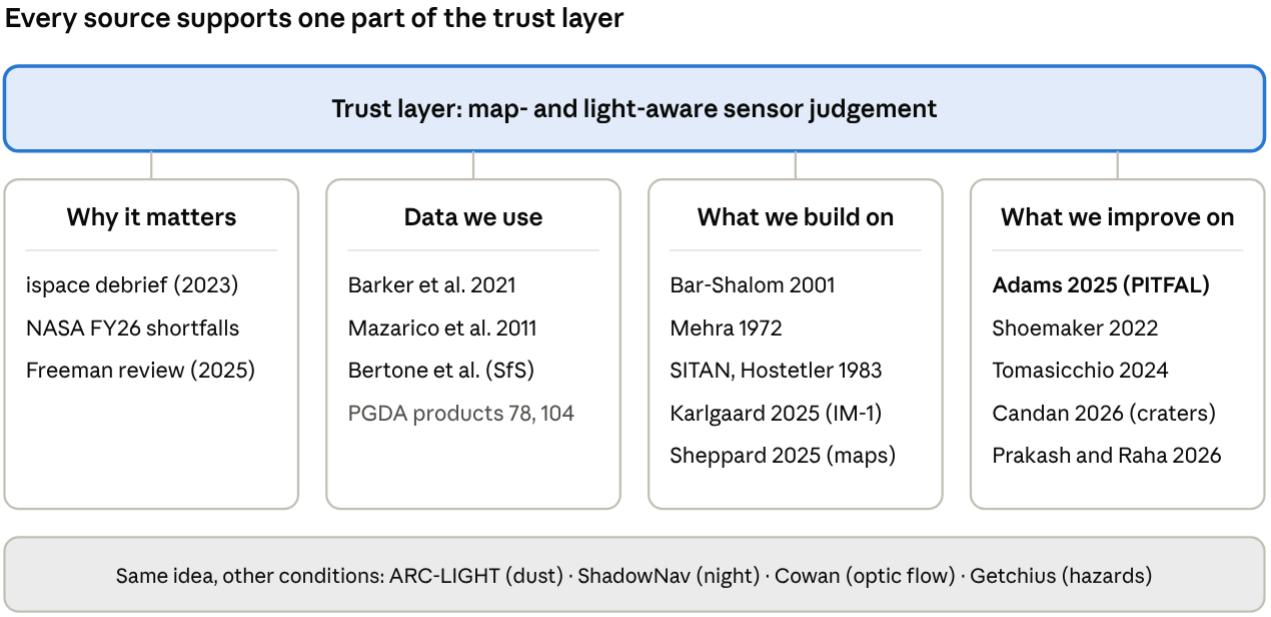
PRJ1 Research Paper Summaries: Sensor Trust for Lunar South Pole Landing

Each research paper used in the project, summarised on its own: the problem it tackles, what it did, what it found, its limits, and how this project uses it. News articles are covered in the Literature Review; only the two key reports appear here.

At a glance



where each source fits · 4 groups plus related work

Adams et al. 2025 (PITFAL) is the closest paper and our main comparison; the data papers supply the terrain and shadows the trust layer reads.

1. Closest prior work

Adams, Ward, Smith, Pedrotty and Mendeck (2025). Topographic slant range modeling and fault detection for precision planetary landing. AIAA SciTech Forum. NTRS 20240014100

- **Problem:** NASA's SPLICE landing system uses a multi-beam lidar (NDL) for range. Modelling range over rough terrain inside an EKF is fragile, especially with noisy terrain maps.

- **What they did:** a new range model that uses the terrain map (DEM) without relying on surface-normal estimates, a fast beam-to-ground intersection method, and PITFAL, a fault check that compares the apparent altitude from each beam.
- **How the fault check works:** each beam's value passes a chi-square test with a fixed limit of 6.635 (99%, one degree of freedom). A beam that fails has its noise inflated until it passes.
- **Tested on:** 1,000 Monte Carlo runs of a closed-loop lunar landing simulation, using a 10 x 10 km patch of the PGDA NPA South Pole map at 5 m.
- **Found:** the filter stayed statistically consistent and met SPLICE's landing-accuracy targets. PITFAL tends to activate on steep, high-gradient terrain.
- **Limits:** the gate is fixed; the response comes after a reading fails; it needs several beams; lighting and camera trust are not covered.
- **Our use:** the main baseline we compare against, and the clearest statement of the gap.

Karlgaard, Lugo, Shidner, Hines, Amzajerdian, Gragossian, Litton and Angster (2025). Navigation Doppler Lidar performance assessment and trajectory reconstruction of the IM-1 lunar landing. NASA TM; AAS/AIAA Astrodynamics Specialist Conference. NTRS 20250010290

- **Problem:** IM-1's own rangefinders never worked, so NASA's experimental NDL lidar became the main sensor. How well did it perform?
- **What they did:** rebuilt the final 3 minutes of IM-1's descent after the flight, combining NDL and IMU data in an Unscented Kalman Filter with a lunar terrain map and ray-tracing of each lidar beam.
- **Found:** the NDL gave valid readings at longer range than predicted; measured and modelled values agreed within about 5 m and 0.5 m/s.
- **Limits:** done after the flight, not during it; does not change how readings are accepted.
- **Our use:** evidence that a terrain-map model of altimeter readings works on real flight data; the ray-tracing idea matches our Unity raycast altimeter.

Prakash D. and Raha (2026). Co-state based data fusion and risk aware filtering for spacecraft navigation and hazard prediction. arXiv 2604.20485 (preprint)

- **Problem:** detecting that a navigation filter is going wrong before it diverges, without predefined fault models or thresholds.
- **What they did:** introduced a "co-state" signal that measures mismatch between measurements and dynamics, and a probabilistic model over behaviour regimes to forecast risk.
- **Tested on:** real lunar powered-descent telemetry (mission not named in the abstract).

- **Found:** inconsistency is detected significantly earlier than an EKF's standard statistical alarms.
- **Limits:** no terrain map or lighting; reacts to inconsistency rather than predicting it; not peer reviewed yet.
- **Our use:** an alternative way to separate "filter wrong" from "sensor wrong" (gap G3).

Sheppard, Reale and Skinner (2025). A stochastic approach to terrain maps for safe lunar landing. arXiv 2512.12058; IEEE Aerospace Conference 2026

- **Problem:** at the shadowed South Pole, vision-based hazard detection is unreliable and lidar is little tested, so landers need better prior maps with honest uncertainty.
- **What they did:** built elevation maps from LRO data with Gaussian processes, where a second model learns how map noise varies across the area from the DEM's confidence data.
- **Found:** the method represents varying sensor noise more accurately and gives more informative terrain uncertainty (no headline numbers in the abstract).
- **Limits:** aimed at hazard detection and site selection, not at the navigation filter.
- **Our use:** supports using the map's own error estimate as part of sensor trust (gap G4).

2. Navigation and lighting

Shoemaker et al. (2022). Terrain relative navigation in a lunar landing scenario using autoNGC. AIAA SciTech 2022. NTRS 20210024481

- **Problem:** testing NASA Goddard's autonomous navigation software (autoNGC) on flight-like hardware for a lunar landing.
- **What they did:** processor-in-the-loop tests combining camera terrain-relative navigation (TRN) and weak-signal GPS.
- **Key detail:** camera fixes are timed so the local Sun elevation is neither too low nor too high: about 20° to 40° between 40 and 15 km altitude.
- **Limits:** lighting is handled by scheduling, not by changing sensor trust.
- **Our use:** shows the standard workaround fails at the South Pole, where the Sun stays at 1.5° to 3°.

Cauligi, Swan, Ono, Daftry, Elliott, Matthies and Atha (2023). ShadowNav: crater-based localization for nighttime and permanently shadowed region lunar navigation. IEEE Aerospace Conference. arXiv 2301.04630

- **Problem:** rovers driving at night or in permanent shadow cannot use normal camera navigation.

- **What they did:** detected crater edges with an external light and a stereo camera, then matched them to known craters from orbital maps.
- **Found:** promising for keeping localisation error under 10 m in simulated night-time imagery.
- **Limits:** a hardware fix (adds a light) for rovers, not landers.
- **Our use:** contrast: we handle shadow in software by lowering camera trust.

Candan and Servadio (2026). Deep learning-based lunar crater terrain relative navigation. arXiv 2606.14776 (preprint)

- **What they did:** a deep-learning crater detector (built for NASA's Crater Detection Challenge), matched to a crater database, with consensus outlier removal and an EKF.
- **Found:** from a 5 km start error, navigation error drops to a few hundred metres in simulation.
- **Limits:** camera noise is a fixed value, unaffected by lighting.
- **Our use:** evidence that current crater navigation ignores lighting in its trust.

Cowan, Fanti, Williams, Yam, Asakuma, Nada and Izzo (2025). Vision-guided optic flow navigation for small lunar missions. arXiv 2511.17720

- **What they did:** estimated motion from camera optical flow plus a laser rangefinder, using flat or spherical terrain models, for low-cost landers.
- **Found:** results over complex South Pole terrain.
- **Limits:** no handling of terrain jumps or shadow.
- **Our use:** shows small missions rely on cheap, simple sensor suites like the one we simulate.

Tomasicchio, Del Prete, Renga, Grassi, Ceresoli, Silvestrini, Lavagna et al. (2024). ION GNSS+ 2024; summarised in Inside GNSS, Apr 2025

- **What they did:** combined camera navigation, IMU, altimeter and signals from planned lunar navigation satellites for a lander.
- **Found:** with four satellites, position errors of tens of metres; outliers removed with a Mahalanobis-distance check.
- **Limits:** fixed outlier gate; altimeter noise fixed at 1% of height; terrain and lighting not modelled.
- **Our use:** shows fixed gates are current practice.

Getchius et al. (2022). Hazard detection and avoidance for the Nova-C lander

- **What they did:** designed Intuitive Machines' onboard hazard detection, pairing precision navigation with machine-vision algorithms.
- **Requirements:** a site clear of hazards over 30 cm with slope under 10°, checked about 400 m above the target within 15 s.
- **Our use:** the source of our 10° tilt and slope limits; the lander family that tipped over on IM-1 and IM-2.

3. Terrain and illumination data

Barker, Mazarico, Neumann, Smith, Zuber and Head (2021). Improved LOLA elevation maps for South Pole landing sites: error estimates and their impact on illumination conditions. *Planetary and Space Science* 203, 105119. NTRS

- **Problem:** small errors in the orbits of the LOLA laser data blur South Pole terrain maps, which matters for landing and for predicting sunlight.
- **What they did:** repeatedly adjusted each laser track to fit the map, reducing orbit-related errors by more than a factor of 10, and produced 5 m/pixel maps with per-pixel error estimates.
- **Found:** horizontal uncertainty of roughly 10–20 cm and vertical 2–4 cm per 16 x 16 km region; map errors measurably affect predicted illumination.
- **Our use:** our terrain source; the per-pixel error maps feed the map-error term in our trust layer. Published as PGDA Product 78, which includes Site04 (Shackleton rim) and Site01 (connecting ridge).

Mazarico, Neumann, Smith, Zuber and Torrence (2011). Illumination conditions of the lunar polar regions using LOLA topography. *Icarus* 211(2), 1066–1081. NTRS

- **What they did:** computed horizon profiles from 240 m LOLA polar maps and simulated sunlight at the poles every hour over a full 18.6-year lunar cycle.
- **Found:** maps of how often each polar location is lit, and where permanent shadow lies.
- **Our use:** the method behind our shadow prediction; it also explains why a 16 km tile is too small at a 2° Sun, since distant terrain casts the shadows.

Bertone, McKenna, Barker, Mazarico, Beyer and Petro. Enhanced topography models for selected lunar South Pole regions with Shape-from-Shading. *The Planetary Science Journal*, doi:10.3847/PSJ/ae5b70. PGDA Product 104

- **Problem:** LOLA laser tracks are sparse at short scales, so fine terrain detail is interpolated.

- **What they did:** used LROC Narrow Angle Camera images and Shape-from-Shading to sharpen LOLA maps of the 13 original Artemis III candidate regions.
- **Found:** 5 m/pixel maps over more than 4,500 km², mostly within 1 m of LOLA reference points, with slope, roughness and illumination products. Data released on Zenodo on 16 Dec 2025.
- **Our use:** optional higher-detail terrain, including a "Connecting Ridge" region that overlaps our Site01.

4. Filtering theory

Bar-Shalom, Li and Kirubarajan (2001). *Estimation with Applications to Tracking and Navigation*. Wiley

- **What it is:** the standard textbook on Kalman filtering for tracking and navigation.
- **Key idea used:** innovation gating. A measurement is accepted only if its normalised innovation (surprise) is below a chi-square threshold chosen for a confidence level, such as 99%.
- **Our use:** this fixed gate is the baseline rule our trust layer replaces with a map-informed one.

Mehra (1972). Approaches to adaptive filtering. *IEEE Transactions on Automatic Control* 17(5), 693–698

- **What it is:** a classic survey of methods that let a Kalman filter re-estimate its own noise levels from incoming data.
- **Limits for our problem:** adaptation happens after residuals grow, so it reacts to a cliff only after the surprise has arrived.
- **Our use:** the reactive approach we contrast with predicting trust from the map in advance.

Hostetler and Andreas (1983). Nonlinear Kalman filtering techniques for terrain-aided navigation. *IEEE Transactions on Automatic Control* 28(3), 315–323

- **What it is:** the SITAN method, which lets an aircraft correct its position with a radar altimeter and a terrain map by linearising the terrain slope under the vehicle in an EKF.
- **Limits for our problem:** a cliff is a jump, not a slope, so the linear approximation breaks exactly there.
- **Our use:** the classic map-plus-altimeter method; our terrain term samples the real height spread instead of a local slope.

5. Related systems and reviews

Cushen, Carrico, Bueno, Wettstein, Adalja, Shi, Garcia and Garcia (2024–25). ARC-LIGHT: Algorithm for Robust Characterization of Lunar surface Imaging for Ground Hazards and Trajectory. NASA Human Lander Challenge 2024 technical paper; *Aerospace* 12(3), 177 (2025)

- **Problem:** in the last ~50 m, engine exhaust throws up dust that blinds cameras and lidar; landers often switch off all sensors except the IMU, and Chang'e 3 saw interference at 20 m.
- **What they did:** built SELENE, a 0.91 x 0.91 x 1.2 m sealed acrylic tank with a downward camera and 2D lidar, fans, and a laser and light sensor to measure how thick the "dust" is. A fine mist of ~1-micron droplets (DEHS) stood in for regolith. They collected hundreds of labelled images, trained a small neural network (100 x 100 image input, three hidden layers of 64 neurons) to estimate dust thickness, and used that estimate to correct lidar distances.
- **Found:** on 20 held-out images the network's estimates matched the measured dust thickness, and corrected lidar scans moved back towards the true geometry. The full design adds a shut-off if errors get too large.
- **Plan:** a 3¼-year path to flight at a $330k baseline cost. Won first place in NASA's 2024 Human Lander Challenge. Team: 3 graduate and 5 undergraduate students; advisors M. Gamba and C. Ruf.
- **Our use:** the same "estimate conditions, then decide how much to trust the sensor" idea, applied to dust; our planned extension.

Freeman (2025). Soft landing on the Moon: the ups and downs of robotic lunar landers. *Journal of Space Operations & Communicator* 21(2)

- **What it is:** a qualitative review of 12 news reports on recent robotic lunar landings.
- **Found:** broken landing legs were the most cited issue (5 of 12), followed by laser altimeter (3), laser rangefinder (2) and laser navigation (2).
- **Limits:** based on news coverage, not technical data.
- **Our use:** background showing sensor problems recur across missions.

6. Key reports

NASA STMD (2026). FY26 Civil Space Shortfall Prioritization

- **What it is:** NASA's ranking of 32 technology shortfalls from community input, plus 40 need statements chosen as STMD's FY26 focus areas.

- **Relevant content:** need 3.02, "Land on lunar south pole exploration sites in various illumination conditions with appropriate accuracy", is listed first of the 40. SF11 (land science payloads) ranks 4th of 32, SF24 (navigation) 8th, SF03 (precise lunar landing) 15th.
- **Our use:** shows the problem is an official NASA priority.

ispace (2023). HAKUTO-R Mission 1 debriefing

- **What it is:** ispace's own analysis of the April 2023 crash, presented by CTO Ryo Ujiie.
- **Found:** the rangefinder started updating altitude from about 15 km. At 16:38 UTC a gap of over 3 km opened at a crater rim and the reading was "not reflected". By 16:43 the lander estimated 0 km while measuring about 5 km; it then ran out of fuel and fell. ispace attributes this to the terrain on the route not being included in simulations, after a landing-site change.
- **Our use:** the clearest real case of correct data rejected by software, and of risk from an untested site change.